

TaxaAssign

Bayesian taxonomic assignment from match scores and priors. Part of the TaxaID ecosystem.

Overview

TaxaAssign makes a consensus taxonomic assignment for each observation. It multiplies the likelihood that a match score corresponds to a particular species (from TaxaLikely) by the prior probability that the species occurs at the sampling location (from TaxaExpect). A key feature is bridging gaps between these two inputs: TaxaAssign generates likelihoods for species that have occurrence records but are missing from the reference database, and generates priors for species that appear in the reference but lack local occurrence records (potential invaders or rare species). After normalization and Monte Carlo simulation, each candidate receives a posterior probability. If a single candidate has strong support, it is assigned as the consensus taxon; otherwise, a coarser rank (genus, family, etc.) is assigned via lowest common ancestor. All competing hypotheses and their probabilities are retained in the output. The user can also update priors iteratively: if one observation strongly supports species A, that evidence can sharpen the prior for species A in other observations from the same sample. By naming plausible species missing from the reference library and assigning them likelihoods, TaxaAssign reduces the false positives that occur when detections of unreferenced species are incorrectly attributed to their closest referenced relative.

Two workflows:

- **Full Bayesian** – TaxaLikely likelihoods + TaxaExpect priors
- **LLM-shortcut** – exponentially-weighted scores + LLM biogeographic priors

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaAssign")
```

Quick Start

LLM-shortcut workflow (simplest)

```
library(TaxaAssign)

out <- run_llm_pipeline(
  match_df      = match_obj,      # from TaxaMatch
  geographic_hint = "Southern California",
  barcode_term   = "12S"
)
head(out$consensus)
```

Full Bayesian workflow

```

out <- run_bayesian_pipeline(
  match_df      = match_obj,          # from TaxaMatch
  model_params  = trained_model,      # from TaxaLikely
  taxaexpect_priors = priors,         # from TaxaExpect
  site = list(grid_id = "Grid_34p1_m119p1", main_habitat = "Marine")
)
head(out$consensus)

```

Step-by-step

```

# 1. Join priors to likelihood output
joined <- join_priors(likelihoods, priors,
  site = list(grid_id = "G1", main_habitat = "Marine"))

# 2. Compute posteriors via Monte Carlo
posteriors <- compute_posterior(joined, n_sims = 1000)

# 3. Consensus taxonomy (LCA among plausible hypotheses)
consensus <- posterior_consensus(posteriors)

# 4. Generate report
report <- generate_report(posteriors, consensus)

```

Key Functions

Core assignment: - `compute_posterior()` – Bayes’ theorem with MC uncertainty - `assign_taxa_llm()` – LLM-shortcut (priors + likelihoods in one call) - `join_priors()` – merge TaxaExpect priors with likelihood output

Consensus: - `posterior_consensus()` – LCA from posterior probabilities - `score_consensus()` – conventional score-based consensus - `update_prior_from_consensus()` – empirical Bayes refinement

Unreferenced species: - `suggest_unreferenced_species()` – LLM-first detection of missing taxa (supports eDNA, acoustic, and image data via `data_type` param) - `expand_unreferenced_hypotheses()` – name generic H2/H3 rows

High-level wrappers: - `run_bayesian_pipeline()` – full Bayesian (TaxaLikely + TaxaExpect -> posteriors) - `run_llm_pipeline()` – LLM-shortcut (match_df -> posteriors in one call)

Context and reporting: - `build_context()` – auto-populate site context from taxon names - `generate_report()` – publication-ready Methods + Results text - `report_assign()` – lightweight section for assemble_report()

Statistical Methods

TaxaAssign computes Bayesian posterior probabilities for each candidate taxonomic assignment:

$$P(H_i | D) = \frac{L(D | H_i) \times \pi(H_i)}{\sum_j L(D | H_j) \times \pi(H_j)}$$

where L is the likelihood (from TaxaLikely or an exponential score-weighting proxy) and π is the prior (from TaxaExpect or LLM estimation).

- **Monte Carlo uncertainty propagation:** priors are modelled as $\text{Beta}(\alpha, \beta)$ distributions and likelihoods as $\text{Normal}(\text{mean}, \text{sd})$; 1000 simulations (default) propagate both sources of uncertainty into posterior means, SDs, and confidence scores (fraction of simulations won)
- **Two workflows:** the full Bayesian workflow uses calibrated likelihoods from TaxaLikely’s hierarchical model and spatially-explicit priors from TaxaExpect’s GLMM; the LLM-shortcut workflow uses exponential score weighting ($L_i = e^{\lambda s_i}$) and LLM-estimated priors with information-quality-driven Beta concentration
- **Posterior consensus:** the smallest set of hypotheses capturing 95% of posterior mass is identified; if multiple taxa remain, the lowest common ancestor (LCA) determines the consensus rank; downranking refines coarse assignments when only one finer-rank taxon exists at the study site
- **Empirical Bayes refinement:** species confidently identified in one observation receive boosted priors in unresolved observations from the same study, analogous to shrinkage estimators (Efron and Morris 1973)
- **Dark diversity fallback:** species absent from TaxaExpect’s spatial model receive priors from Tier 3 (undetected species) estimates, preventing false negatives from incomplete occurrence data

For the full statistical derivation, assumptions, and references, see `inst/TaxaAssign_supplemental_methods.md`.

Vignettes

- Taxonomic Assignment – full workflow
- TaxaID Ecosystem Overview – cross-package guide

Part of TaxaID

TaxaAssign is the convergence point of the TaxaID ecosystem. It receives likelihoods from TaxaLikely and priors from TaxaExpect, then passes assignments to TaxaFlag for quality screening.

Ecosystem: TaxaLikely + TaxaExpect -> **TaxaAssign** -> TaxaFlag

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- TaxaTools (foundation package)
- TaxaLikely and TaxaExpect (for the Bayesian workflow; in Suggests)
- An LLM API key is needed for the LLM-shortcut workflow

All dependencies are declared in the DESCRIPTION file and installed automatically.

Developed with Claude Code (Anthropic).